ADS508 Data Science With Cloud Computing

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Company Industry: Real Estate Analytics
Company Size: 2

Abstract:
J&F Realty Insighters is a real estate analytics company that provides data driven insights into housing market trends. Currently facing the challenge of accurately identifying the factors that influence housing price trends, this project will develop a machine learning model that predicts housing price trends by analyzing housing supply, demand indicators, and historical home values.

Problem Statement:
Real estate investment has traditionally relied on historical trends, expert intuition, and manual data analysis. However, the housing market is a high-stakes environment shaped by a complex web of variables—from hyper-local demographics to broad economic shifts. Without a rigorous, data-driven framework, investors risk inaccurate valuations and missed opportunities in emerging markets.
J&F Realty Insighters bridges this gap. By utilizing machine learning to synthesize massive housing and demographic datasets, we provide precise property value estimations and identify the hidden patterns driving market growth. Our predictive models empower investors to make informed, high-conviction decisions while significantly mitigating financial risk.

Goals:
- Predict: Deliver precision property valuations through advanced machine learning.
- Analyze: Pinpoint the key demographic and economic factors driving home prices.
- Scale: Leverage AWS to build a robust, cloud-native data architecture.
- Identify: Help investors discover the next high-growth real estate markets.
- Optimize: Prove that data-driven insights outperform manual analysis in property investment.

Non-Goals:
- Financial Advice: This project provides data analysis, not personalized investment or financial counseling.
- Real-Time Data: Predictions are based on historical datasets rather than live, real-time market feeds.

- Production Deployment: We will not be integrating the model into a commercial-grade real estate platform.
- Proprietary Access: Integration with private or third-party proprietary databases is excluded.
- Short-Term Volatility: Our focus is on long-term trends rather than daily or weekly price fluctuations.

Data Sources [https://www.zillow.com/research/data/]:

- Zillow Home Value Index (ZHVI): Represents the estimated typical home value and serves as the primary target variable for our predictive model.
- Median Sale Price: Provides historical transaction data to validate model accuracy against actual market sales.
- Market Temperature Index: Measures supply-demand competitiveness to identify "hot" seller's markets.
- Housing Inventory: Tracks available for-sale stock to quantify the supply-side impact on price fluctuations.

Technical Workflow

The data architecture is built for scale and reproducibility using the AWS ecosystem:

- Storage: Raw historical time-series data is staged in Amazon S3.
- Ingestion: Data is pulled into AWS SageMaker Studio for exploratory analysis and development.
- Transformation: We convert datasets from wide to long format, merging them via shared geographic identifiers (Region/State) to create a unified feature set.
- Modeling: This integrated dataset powers a regression model designed to forecast price trends based on supply and demand dynamics.

Data Exploration:

This project utilizes Amazon S3 for centralized data storage, selected for its scalability, durability, and cost-effectiveness. Using Amazon S3 ensured the integration with the Amazon SageMaker smoothly. The Zillow housing datasets were uploaded via the AWS Management Console to a dedicated bucket, establishing a cloud foundation for the project's dataset. By referencing the data through unique S3 URLs, workflow of manual downloading was eliminated, keeping the workflow within the AWS environment.

- Tools that will be used to ingest and explore data.
    - Amazon S3
        - Central storage location for the datasets. Allows the data to be accessed directly by SageMaker without local file storage.
    - Amazon SageMaker Studio

- Primary development environment for this project. Amazon SageMaker Studio provides a jupyter notebook interface that allows data scientists at J&F Realty Insighters to run Python codes, access S3 data, and perform any necessary data analysis in the cloud.
  - Python
    - Python was selected, as it is widely used in data science workflows.

During the data exploration phase, the primary goal is to understand the structure and quality of the datasets before building predictive models. We used commands like head(), shape, and columns to see the structure of the Zillow Home Value Index dataset. Most of the columns represent monthly home value data, while the first few columns contain information about each metro area such as region name and state. We also ran describe() to look at summary statistics and used isnull().sum() to check for missing values. Overall, the dataset looked fairly complete with only a small number of missing entries.

We then created a simple bar chart to visualize some metro areas with higher home value index values. This helped us quickly see that housing prices vary across regions. At this stage the goal was mainly to understand the dataset and confirm that it could be used to analyze housing trends over time.

Data Ingestion and Exploration Code

The following example code was implemented in a SageMaker Studio notebook to ingest and explore the Zillow dataset stored in Amazon S3

In [2]:
```
import pandas as pd
```

In [3]:
```
df = pd.read_csv("s3://ads508-housing-data-faye/Metro_zhvi_uc_sfrcondo_tier_0.33_0.67_sm_sa_month.csv")
df.head()
```

Out[3]:

	RegionID	SizeRank	RegionName	RegionType	StateName	2000-01-31	2000-02-29	2000-03-31	2000-04-30	2000-05-31	...	2025-04-30	2025-05-31	2025-06-30	2025-07-31	2025-08-3
0	102001	0	United States	country	NaN	120245.401273	120456.951481	120719.304889	121282.001451	121929.597731	...	356520.996883	355847.084537	355240.355926	354820.996430	354597.24283
1	394913	1	New York, NY	msa	NY	216213.074441	217131.858016	218059.151183	219938.205956	221884.032245	...	684527.238403	686280.002733	687742.249805	688904.163124	689687.00471
2	753899	2	Los Angeles, CA	msa	CA	219357.633964	220173.922681	221261.211467	223424.550904	225790.566038	...	943189.059655	938463.706961	934153.202642	931587.011384	930526.40611
3	394463	3	Chicago, IL	msa	IL	149975.869412	150114.703378	150379.115111	151036.905792	151828.148861	...	324141.432142	324425.469135	324778.263087	325598.611099	326571.86493
4	394514	4	Dallas, TX	msa	TX	126453.868825	126510.191820	126574.940868	126743.087346	126964.783873	...	367625.466282	365231.672338	362889.504974	360891.495428	359541.99378

5 rows × 318 columns

Link to the code used for ingestion and exploration: https://github.com/jsryu-git/ADS508_Final/blob/22caef8fb4a463c3028bae6f7dbea27615255b0d/ads508_module3_faye_jun.ipynb

Data Preparation:

Commented [JR1]: Module 4

How will you transform the raw data so that it is ready for training?

- Data Scrubbing: What data cleansing techniques will you apply?
- Feature Selection: Which fields from your data will you use/exclude?
- Feature Creation: Which fields will be combined, or bucketed?
- Feature Transformation: What other transformations (such as one hot encoding, etc.) will you apply to your data?

- How will you balance your data set?
- How will you split your dataset?

Model Training:

Commented [JR2]: Module 5

How will you train your model, what tools will you use?

- Will you be using SageMaker Jumpstart, built-in-algorithms, bring-your-own-script or bring-your-own container?
- Which algorithm will you be using?
- Which parameters will you be passing?
- Which instance size/count will you be using?
- How will you evaluate your model?

Measuring Impact:

Two key metrics will be used to evaluate the impact of this project.

Prediction Accuracy (Root Mean Squared Error)
The first metric is Root Mean Squared Error (RMSE), which measures the difference between predicted housing prices and the actual values in the dataset. A lower RMSE indicates that the predictive model is accurately capturing housing price trends.

Model Performance (R^2 Score)
The second metric is the R^2 score, which measures how much variance in housing prices is explained by the model. A higher R^2 score indicates that the model successfully captures the relationship between housing supply, demand indicators, and home values.

Together, these metrics provide a quantitative way to evaluate whether the predictive model improves housing price analysis compared to traditional manual analysis.

Security Checklist, Privacy and Other Risks:

The datasets used in this project do not contain protected health information or personal identifiable information. The Zillow datasets represent aggregated housing market data and do not include records tied to individual homeowners or buyers.

This project will not track user behavior or store financial information such as credit card data. All datasets are stored in an Amazon S3 bucket and accessed through SageMaker notebooks for analysis.

The application reads data from the following S3 bucket: s3://ads508-housing-data-faye

Commented [JR3]: Temporary. Will be changed to specific bucket for this project

One potential data bias to consider is uneven geographic representation. Some metropolitan areas may have longer or more complete historical data records than others. This imbalance could cause predictive models to perform better in highly represented regions while underperforming in areas with limited data.

From an ethical perspective, predictive housing models must be interpreted carefully. Housing market predictions could influence investment decisions and potentially contribute to speculation or price inflation in certain areas. Because of this, the model should be presented as an analytical tool rather than financial advice.

Future Enhancements:

Commented [JR4]: Module 6

Provide at least 3 ways you would improve your model pipeline if you had more time/additional resources?